

# Lesson 07 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

**Provenance:** All 25 items are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

- Q1. B.** RAM stands for Random Access Memory. A, C, and D are not expansions of the abbreviation.
- Q2. C.** ROM stands for Read-Only Memory. A, B, and D are not the correct full forms.
- Q3. A.** Speed decreases from registers to cache to main memory (RAM) to secondary storage. B swaps the top two levels; C and D scramble the order.
- Q4. D.** Registers are the fastest storage in the computer. L1 cache (B), RAM (A), and an SSD (C) are each progressively slower.
- Q5. B.** Cache is faster than main memory. It is slower than RAM is false (A); cache is smaller, not larger (C); and cache is more expensive per bit, not cheaper (D).
- Q6. C.** Cache is smaller and faster than RAM, while RAM is larger and slower. A reverses the size relation; B claims they are identical; D wrongly calls cache a backup.
- Q7. A.** L1 is the smallest and fastest cache level. L2 and L3 are successively larger and slower; L4 is not a standard cache level.
- Q8. D.** L3 is the largest and slowest of the three cache levels, though still faster than main memory. L1 (A) is smallest and fastest; L2 (B) is the middle level; registers (C) are not a cache level.
- Q9. C.** Cache is not a permanent backup; it holds copies of data that live in RAM. Storing recent data (A), exploiting locality (B), and being faster than main memory (D) are all true.
- Q10. B.** RAM is volatile and ROM is non-volatile. A reverses the pair; C describes ROM as read-only and RAM as writable instead of the volatility distinction; D misclassifies both as storage types.
- Q11. A.** RAM is volatile and loses its contents when the power is switched off. ROM (B), the hard disk drive (C), and the solid-state drive (D) are all non-volatile.
- Q12. D.** RAM is primary memory, directly accessible by the CPU. An SSD (A), an HDD (B), and magnetic tape (C) are secondary or tertiary storage, not primary memory.
- Q13. C.** A solid-state drive is secondary storage. A register (A), cache (B), and RAM (D) are primary (CPU-side or main memory), not secondary storage.

- Q14. A.** The hierarchy exists because faster memory is smaller and more expensive per bit, so a layered arrangement balances speed, cost, and capacity. B claims all memory is equal; C names the slowest level as fastest; D overstates the size of registers.
- Q15. B.** Statements I and II are correct: registers are the fastest storage, and cache is faster than main memory. III is false because RAM is volatile. A omits II; C includes false III and omits I; D includes false III.
- Q16. A.** Reusing recently accessed data and nearby data is locality of reference. Overflow (B), fragmentation (C), and deadlock (D) are unrelated concepts.
- Q17. C.** Temporal locality is reusing the same data or instruction again soon after it was accessed. A describes spatial locality; B describes persistence on storage; D describes volatility.
- Q18. D.** Registers are the smallest storage. L1 cache (C) is the next smallest; RAM (B) and an SSD (A) are larger.
- Q19. B.** Secondary storage such as an SSD or HDD is non-volatile and keeps files between sessions. Registers (A), cache (C), and RAM (D) do not retain their contents without power.
- Q20. D.** An SSD is secondary storage that retains data without power, so this pairing is correct. A mislabels registers as slowest (they are fastest); B wrongly calls cache a non-volatile backup; C wrongly calls RAM non-volatile and read-only.
- Q21. A.** Registers sit at the top of the hierarchy and are fastest. B, C, and D are slow secondary media.
- Q22. A.** Speed falls from registers to cache to RAM to hard disk. B, C, and D scramble that order.
- Q23. A.** Cache bridges the CPU–RAM speed gap. Tape (B), floppy (C), and printer hardware (D) do not.
- Q24. A, B, C.** RAM, ROM, and cache are primary memory. The hard disk (D) is secondary storage.
- Q25. A, B, C.** Register speed, cache behaviour, and secondary-storage persistence are correct. D is false: ROM is non-volatile.
- Q26. A.** Cache sits between CPU and RAM to reduce access time. HDDs, optical discs, and tape are secondary storage and are slower.
- Q27. A.** Registers are closest to the CPU and fastest. RAM, SSD, and tape are progressively slower storage levels.
- Q28. B.** Larger, cheaper storage is generally slower, which motivates the hierarchy. Not all levels are volatile, and memory is not removed from the CPU.
- Q29. B.** The usual order is registers, cache, then RAM. The other options reverse one or more levels.
- Q30. A, B, C.** Registers are small/fast, cache reduces access time, and secondary storage retains data without power. Tape is much slower than registers.